

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

*I **C. PRIYANKA (192511074)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. PRIYANKA

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

ANSWERS:

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive Learning

Inductive learning is a machine learning approach in which the model learns general patterns from previously observed patient data and uses those patterns to predict the outcome for new patients.

For this healthcare scenario, a **supervised inductive learning approach** is suitable because the historical patient records contain input features along with a known diabetes diagnosis.

Target Variable

The target variable is:

Diabetes Diagnosis

Target Value	Meaning
0	Non-Diabetic
1	Diabetic

The model learns the relationship between patient characteristics and diabetes diagnosis.

Important Features

Feature	Importance / Justification
Glucose Level	One of the strongest indicators of diabetes because abnormal blood glucose is directly associated with diabetes.
BMI	High BMI/obesity is strongly associated with increased risk of Type diabetes.
Blood Pressure	High blood pressure can be associated with metabolic disorders and increased diabetes risk.
Age	Diabetes risk generally increases with age, making age useful for prediction.
Family History	A family history of diabetes indicates genetic and hereditary risk.
Cholesterol	Abnormal cholesterol levels can provide additional information about metabolic health.

Inductive Learning Process

Historical Patient Records



Data Preprocessing



Feature Selection



Training Data



Decision Tree Learning



Learn General Patterns



New Patient Data



Diabetes Prediction

For example, the model may learn that:

High glucose + high BMI + positive family history → high probability of diabetes.

Thus, the system generalises patterns from known patients to diagnose new patients.

1(b) Handling Class Imbalance

The dataset may contain considerably more **non-diabetic patients than diabetic patients**.

For example:

Non-Diabetic → 80%

Diabetic → 20%

If the model is trained directly on this imbalanced dataset, it may become biased toward the majority class.

For example, a model predicting "Non-Diabetic" for almost every patient could achieve high accuracy while missing many diabetic patients.

Recommended Strategy: SMOTE

I would use **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE generates synthetic examples of the minority class rather than simply duplicating existing diabetic records.

Process

Original Training Data



Identify Minority Class

(Diabetic Patients)



Apply SMOTE



Generate Synthetic

Diabetic Samples



Balanced Training Data



Train Decision Tree

Why SMOTE is suitable

SMOTE is appropriate because:

1. It increases the number of diabetic training samples.
2. It reduces bias toward non-diabetic patients.

3. It helps the classifier learn the characteristics of diabetic patients.
4. It can improve **Recall/Sensitivity**, which is extremely important in diabetes screening.
5. It avoids simply duplicating the same minority samples.

Important: SMOTE should be applied **only to the training data**, not before the train-test split. Otherwise, synthetic information can leak into the test set and produce misleadingly high performance.

Comparison of Strategies

Strategy	Advantage	Limitation
SMOTE	Creates synthetic minority samples	Can create noisy/overlapping samples
Under-sampling	Simple and fast	May remove useful non-diabetic information
Class Weights	No synthetic data required	May not fully solve severe imbalance
Recommended: SMOTE	Improves minority-class learning	Must be applied only to training data

1(c) 70:30 Dataset Split and Preprocessing

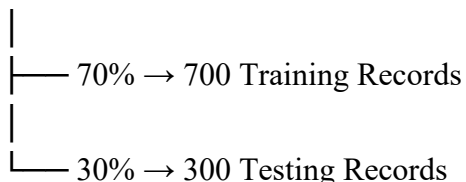
Train-Test Split

The dataset is divided into:

- **70% → Training data**
- **30% → Testing data**

Example:

1000 Patient Records



Why 70:30?

The 70:30 split provides enough data for the model to learn patterns while retaining a sufficiently large independent test set for reliable evaluation.

For a medical application, it is important that the test data remains **unseen during training** so that the model's ability to generalise to new patients can be evaluated.

A **stratified split** should be used so that approximately the same diabetic/non-diabetic proportion is maintained in both training and testing sets.

Preprocessing Steps

1. Handling Missing Values

Missing values should be identified and handled.

For numerical features:

- Glucose
- BMI
- Blood Pressure

- Cholesterol
- Age

Median imputation can be used because it is less affected by extreme values.

2. Encoding Categorical Data

Family History can be encoded as:

No → 0

Yes → 1

The target variable can similarly be represented as:

Non-Diabetic → 0

Diabetic → 1

3. Normalisation

Numerical variables can be standardised using:

where:

- = original value
- = mean
- = standard deviation

This puts numerical features on a comparable scale.

However, for a Decision Tree, normalisation is not strictly necessary, because Decision Trees make decisions using threshold-based splits and are generally insensitive to feature scaling.

Overall Preprocessing

Raw Patient Data



Handle Missing Values



Encode Family History



70:30 Stratified Split



SMOTE on Training Data



Normalise Numerical Features



Training + Testing Dataset

TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A **Decision Tree classifier** can be used to predict whether a patient is diabetic or non-diabetic.

The tree selects the feature that provides the best separation between diabetic and non-diabetic patients.

Two common criteria are:

Information Gain

Higher Information Gain means the feature provides better separation.

Gini Index

A lower Gini value indicates a purer node.

Root Node Selection

For this healthcare scenario, **Glucose Level** would typically be expected to be the root node because it has a strong relationship with diabetes diagnosis.

Illustrative Feature Evaluation

Feature	Information Gain	Gini Index After Split
Glucose	0.241	0.286
BMI	0.137	0.351
Age	0.094	0.382
Blood Pressure	0.071	0.401
Family History	0.063	0.409
Cholesterol	0.041	0.421

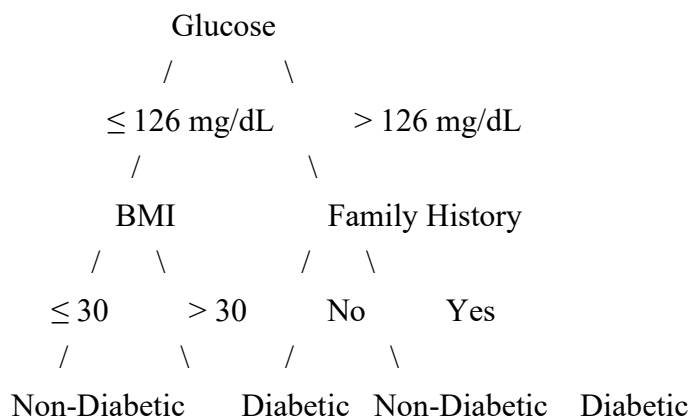
Therefore:

Glucose has the highest Information Gain and lowest Gini Index among the candidate features, so it is selected as the root node.

These numerical values are illustrative because no actual dataset was supplied.

First Three Levels of Decision Tree

A possible tree structure for the given scenario is:



Interpretation

- **Level 1:** Glucose is the root.
- **Level 2:** BMI and Family History provide further separation.
- **Level 3:** The model reaches a prediction based on the patient's combined characteristics.

For example:

Patient A

Glucose = 160

BMI = 32

Family History = Yes



High glucose



High-risk characteristics



Predicted: Diabetic

This is an example tree structure; actual thresholds should be obtained from the trained dataset.

2(b) Model Evaluation

The Decision Tree should be evaluated using:

1. Accuracy
2. Precision
3. Recall
4. F1-Score

Because this is a medical diagnosis problem, **Recall is particularly important.**

Illustrative Confusion Matrix

Assume the test set contains 150 patients:

	Predicted Non-Diabetic	Predicted Diabetic
Actual Non-Diabetic	84 (TN)	16 (FP)
Actual Diabetic	7 (FN)	43 (TP)

Where:

- **TP = 43** → correctly identified diabetic patients
- **TN = 84** → correctly identified non-diabetic patients
- **FP = 16** → non-diabetic patients incorrectly classified as diabetic
- **FN = 7** → diabetic patients incorrectly classified as non-diabetic

Accuracy

Accuracy = 84.67%

Precision

Precision = 72.88%

This indicates that approximately 72.88% of patients predicted as diabetic were actually diabetic.

Recall

Recall = 86%

This means the model correctly detects approximately 86% of the actual diabetic patients.

F1-Score

F1-Score = 78.18%

Evaluation Summary

Metric	Illustrative Result
Accuracy	84.67%
Precision	72.88%
Recall	86.00%
F1-Score	78.18%

False Negatives

A **False Negative (FN)** occurs when:

The patient is actually diabetic, but the model predicts the patient as non-diabetic.

In the example:

FN = 7 patients

This is particularly dangerous in a healthcare system.

Why False Negatives Are Important

A missed diabetic patient may:

- remain undiagnosed,
- delay medical consultation,
- continue unhealthy lifestyle habits,
- develop complications,
- miss early treatment or monitoring.

Therefore, reducing false negatives is clinically important.

How to Reduce False Negatives

The system should prioritise **high recall/sensitivity**.

Possible methods include:

1. Use SMOTE

Increase representation of diabetic patients in the training data.

2. Use Class Weights

Give a higher penalty to incorrectly classifying diabetic patients.

For example:

Non-Diabetic → Weight 1

Diabetic → Weight 2 or higher

3. Adjust Decision Threshold

The classification threshold can be adjusted to classify more borderline patients as high-risk.

4. Use Recall as an Important Evaluation Metric

Instead of optimising only accuracy, the model should consider:

Recall + F1-Score + Specificity + Clinical safety.

Clinical Recommendation

For a screening system, it is generally preferable to tolerate some **False Positives** rather than miss diabetic patients, because patients identified as high-risk can subsequently undergo confirmatory clinical testing.

2(c) Post-Pruning Using Cost-Complexity Pruning

A fully grown Decision Tree may become very complex and learn noise from the training data.

This is called **overfitting**.

For example:

Very Deep Tree



Learns training data extremely well



Poor generalisation



Unreliable predictions on new patients

Therefore, pruning is required.

Cost-Complexity Pruning

Cost-complexity pruning balances:

- prediction error
- tree complexity

The objective can be represented as:

where:

- = classification error
- = complexity parameter
- = number of terminal nodes

Increasing results in a simpler tree.

Pre-Pruning vs Post-Pruning

Pre-Pruning

The tree is restricted while it is being created.

Parameters can include:

- max_depth
- min_samples_split
- min_samples_leaf

Example:

Maximum depth = 4

Minimum samples per leaf = 10

Post-Pruning

First, a relatively large tree is created and then unnecessary branches are removed using cost-complexity pruning.

Full Decision Tree



Calculate pruning cost



Remove unnecessary branches



Simpler Decision Tree



Evaluate on Test Data

Illustrative Comparison

Model	Accuracy	Recall	F1-Score	Tree Complexity
Pre-Pruned Tree	82.00%	80.00%	76.19%	Lower
Post-Pruned Tree	84.67%	86.00%	78.18%	Moderate
Unpruned Tree	80.00%	88.00%	76.52%	Very High

These are **illustrative values**, not results calculated from an actual dataset.

Which Model Is Appropriate for Clinical Deployment?

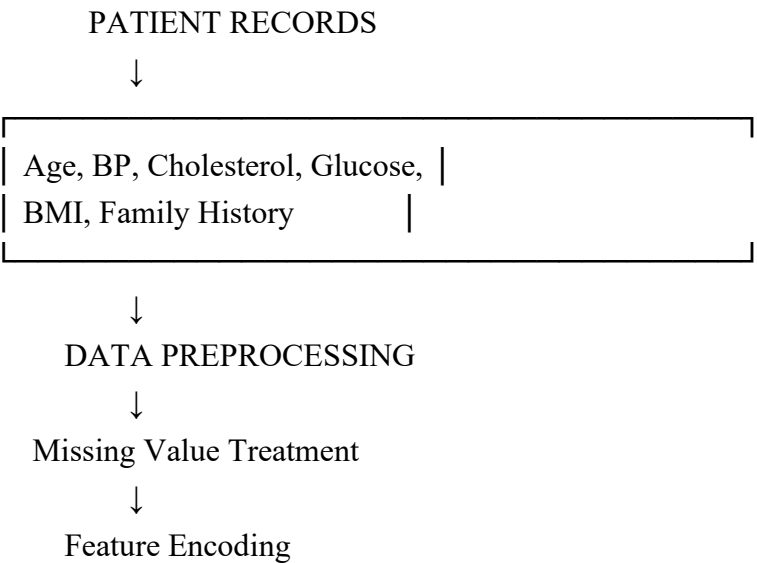
The **post-pruned Decision Tree** is more appropriate for deployment in this scenario, provided that validation on an independent clinical dataset confirms its performance.

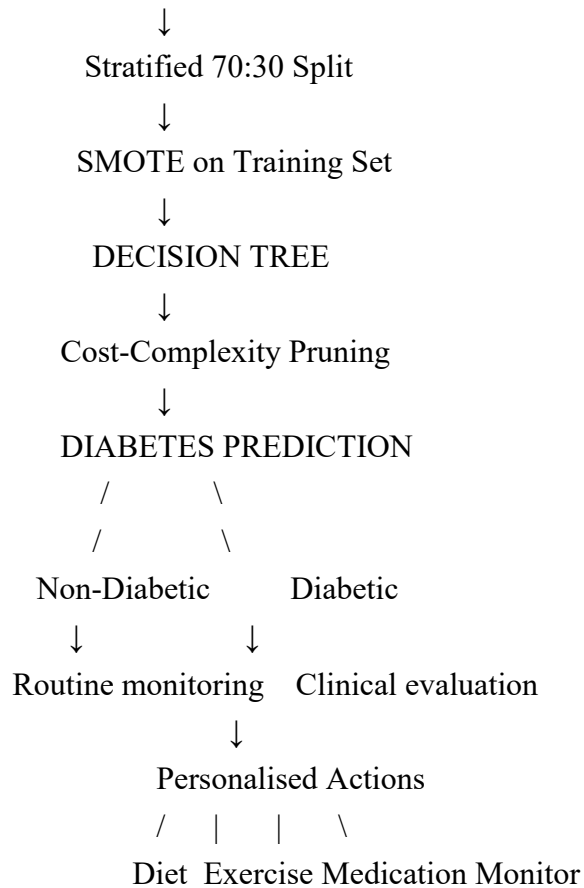
Reasons

- 1. **Reduces** **overfitting**
Unnecessary branches are removed.
- 2. **Better** **generalisation**
The model is more likely to perform consistently on unseen patients.
- 3. **Interpretability**
A smaller tree is easier for doctors and healthcare staff to understand.
- 4. **Improved** **reliability**
A simpler model is less likely to depend on random patterns or noise.
- 5. **Good** **recall**
The illustrative post-pruned model maintains a high recall of 86%, which is important for identifying diabetic patients.
- 6. **Clinical** **explainability**
The system can explain decisions using factors such as glucose, BMI, age, and family history rather than providing only a black-box prediction.

Overall System for the Healthcare Company

The complete solution can be represented as:





Final Conclusion

The healthcare company should use **supervised inductive learning** to learn patterns from historical patient records and predict diabetes for new patients. **Glucose, BMI, blood pressure, age, and family history** are important predictive features. Since diabetic cases are likely to be the minority class, **SMOTE on the training data** can reduce class imbalance. A **stratified 70:30 train-test split** provides sufficient data for learning while maintaining an independent test set.

A **Decision Tree** is suitable because it is interpretable and can show how patient characteristics contribute to the diagnosis. **Glucose level** would typically be a strong root feature based on Information Gain/Gini evaluation. For clinical screening, **False Negatives must be minimised**, so recall should receive greater importance than accuracy alone. Finally, **cost-complexity post-pruning** can reduce overfitting and produce a simpler, more interpretable model suitable for clinical decision support, subject to proper clinical validation before real-world deployment.

